



Analisi e Progetto di Algoritmi

Laurea Triennale in Informatica
(codice: E3101Q113)

AA 2025/2026

**LCS in cui sono presenti
due elementi rossi consecutivi**

Docente: Raffaella Rizzi

[Problema LCS2red(X,Y)]

INPUT:

- $X = \langle x_1, x_2, \dots, x_m \rangle$
- $Y = \langle y_1, y_2, \dots, y_n \rangle$
- $\text{col} : \Sigma \rightarrow \{\text{"red"}, \text{"green"}, \text{"blue"}\}$

OUTPUT:

- **LCS2red(X,Y)**, Longest Common Subsequence in cui sono presenti due elementi rossi consecutivi
(**NB**: la sequenza vuota non è una soluzione)

[Primo sottoproblema ausiliario]

Sottoproblema ausiliario di dimensione $(i, j, 1)$

Trovare $\text{LCS2red}_v(X_i, Y_j)$, cioè la LCS dei prefissi X_i e Y_j , in cui sono presenti due elementi rossi consecutivi, vincolata a terminare con l'elemento x_i se $x_i = y_j$. Se $x_i \neq y_j$, allora $\text{LCS2red}_v(X_i, Y_j)$ non esiste.

$$i \in \{1, \dots, m\}$$

$$j \in \{1, \dots, n\}$$

$$c_{ij1} = |\text{LCS2red}_v(X_i, Y_j)|$$

Numero totale di sottoproblemi: mn

[Secondo sottoproblema ausiliario]

Sottoproblema ausiliario di dimensione $(i, j, 0)$

Trovare $\text{LCS0red}_v(X_i, Y_j)$, cioè la LCS dei prefissi X_i e Y_j , in cui non sono presenti due elementi rossi consecutivi, vincolata a terminare con l'elemento x_i se $x_i = y_j$. Se $x_i \neq y_j$, allora $\text{LCS0red}_v(X_i, Y_j)$ non esiste.

$$i \in \{1, \dots, m\}$$

$$j \in \{1, \dots, n\}$$

$$c_{ij0} = |\text{LCS0red}_v(X_i, Y_j)|$$

Numero totale di sottoproblemi: mn

Soluzione ottimale e valore ottimo

Soluzione ottimale

$$\text{LCS2red}(X,Y) \equiv \max\{\text{LCS2red}_v(X_i, Y_j) \mid 1 \leq i \leq m, 1 \leq j \leq n\}$$

$$\text{Valore ottimo} \equiv \max\{c_{ij1} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$$

[Eq. ricorrenza – Casi base]

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i \neq y_j$$

$$\Rightarrow \nexists \text{ LCS2red}_v(X_i, Y_j)$$

$$\Rightarrow c_{ij1} = -\infty$$

$$\Rightarrow \nexists \text{ LCS0red}_v(X_i, Y_j)$$

$$\Rightarrow c_{ij0} = -\infty$$

Eq. ricorrenza - Passo ricorsivo per $\text{LCS2red}_v(X_i, Y_j)$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) = \text{"red"}$$

$$\text{LCS2red}_v(X_i, Y_j) = \langle \dots, x_i \rangle$$

CASO1: $\text{LCS2red}_v(X_i, Y_j)$ finisce con due elementi consecutivi rossi (di cui x_i è il secondo) e prima non ci sono altri elementi consecutivi rossi.

$$\text{LCS2red}_v(X_i, Y_j) = \langle \dots, x_p, x_i \rangle = \text{LCS0red}_v(X_p, Y_q) + \langle x_i \rangle$$

$$\text{LCS0red}_v(X_p, Y_q) = \max\{\text{LCS0red}_v(X_h, Y_k) \mid 1 \leq h < i, 1 \leq k < j, \text{col}(x_h) = \text{"red"}\}$$

$$\max\{\emptyset\} \Rightarrow \nexists \text{LCS0red}_v(X_p, Y_q) \Rightarrow \nexists \text{LCS2red}_v(X_i, Y_i)$$

$$c_{ij1} = c_{pq0} + 1$$

$$c_{pq0} = \max\{c_{hk0} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j, \text{col}(x_h) = \text{"red"}\}$$

$$c_{ij1} = \max\{c_{hk0} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j, \text{col}(x_h) = \text{"red"}\} + 1$$

$$\max\{\emptyset\} \Rightarrow c_{ij1} = -\infty$$



Eq. ricorrenza - Passo ricorsivo per $\text{LCS2red}_v(X_i, Y_j)$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) = \text{"red"}$$

$$\text{LCS2red}_v(X_i, Y_j) = \langle \dots, x_i \rangle$$

CASO2: $\text{LCS2red}_v(X_i, Y_j)$ finisce (o non finisce) con due elementi consecutivi rossi e prima ci sono elementi consecutivi rossi.

$$\text{LCS2red}_v(X_i, Y_j) = \langle \dots, x_p, x_i \rangle = \text{LCS2red}_v(X_p, Y_q) + \langle x_i \rangle$$

$$\text{LCS2red}_v(X_p, Y_q) = \max\{\text{LCS2red}_v(X_h, Y_k) \mid 1 \leq h < i, 1 \leq k < j\}$$

$$\max\{\emptyset\} \Rightarrow \nexists \text{LCS0red}_v(X_p, Y_q) \Rightarrow \nexists \text{LCS2red}_v(X_i, Y_i)$$

$$c_{ij1} = c_{pq1} + 1$$

$$c_{pq1} = \max\{c_{hk1} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\}$$

$$c_{ij1} = \max\{c_{hk1} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\} + 1$$

$$\max\{\emptyset\} \Rightarrow c_{ij1} = -\infty$$

Eq. ricorrenza - Passo ricorsivo per $\text{LCS2red}_v(x_i, y_j)$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) = \text{"red"}$$

CASO1:

$$c_{ij1} = \max\{c_{hk0} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j, \text{col}(x_h) = \text{"red"}\} + 1$$
$$\max\{\emptyset\} \Rightarrow c_{ij1} = -\infty$$

CASO2:

$$c_{ij1} = \max\{c_{hk1} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\} + 1$$
$$\max\{\emptyset\} \Rightarrow c_{ij1} = -\infty$$

Eq. ricorrenza - Passo ricorsivo per $\text{LCS2red}_v(x_i, y_j)$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) = \text{"red"}$$

$$c_{ij1} = \max \left\{ \begin{array}{l} \max\{c_{hk0} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j, \text{col}(x_h) = \text{"red"}\} + 1 \\ \max\{\emptyset\} \Rightarrow c_{ij1} = -\infty \\ \max\{c_{hk1} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\} + 1 \\ \max\{\emptyset\} \Rightarrow c_{ij1} = -\infty \end{array} \right.$$



Eq. ricorrenza - Passo ricorsivo per $\text{LCS2red}_v(X_i, Y_j)$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) \neq \text{"red"}$$

$$\text{LCS2red}_v(X_i, Y_j) = \langle \dots, x_i \rangle$$

$\text{LCS2red}_v(X_i, Y_j)$ non finisce con due elementi consecutivi rossi e ci devono essere quindi altri elementi consecutivi rossi.

$$\text{LCS2red}_v(X_i, Y_j) = \langle \dots, x_p, x_i \rangle = \text{LCS2red}_v(X_p, Y_q) + \langle x_i \rangle$$

$$\text{LCS2red}_v(X_p, Y_q) = \max\{\text{LCS2red}_v(X_h, Y_k) \mid 1 \leq h < i, 1 \leq k < j\}$$

$$\max\{\emptyset\} \Rightarrow \nexists \text{LCS0red}_v(X_p, Y_q) \Rightarrow \nexists \text{LCS2red}_v(X_i, Y_i)$$

$$c_{ij1} = c_{pq1} + 1$$

$$c_{pq1} = \max\{c_{hk1} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\}$$

$$c_{ij1} = \max\{c_{hk1} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\} + 1$$

$$\max\{\emptyset\} \Rightarrow c_{ij1} = -\infty$$



Eq. ricorrenza - Passo ricorsivo per $\text{LCS2red}_v(x_i, y_j)$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) \neq \text{"red"}$$

$$c_{ij1} = \max\{c_{hk1} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\} + 1$$

$$\max\{\emptyset\} \Rightarrow c_{ij1} = -\infty$$

Eq. ricorrenza - Passo ricorsivo per $\text{LCS2red}_v(x_i, y_j)$

In conclusione, per $\text{LCS2red}_v(x_i, y_j)$:

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) \neq \text{"red"}$$

$$c_{ij1} = \max\{c_{hk1} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\} + 1$$
$$\max\{\emptyset\} \Rightarrow c_{ij1} = -\infty$$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) = \text{"red"}$$

$$c_{ij1} = \max \left\{ \begin{array}{l} \max\{c_{hk0} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j, \text{col}(x_h) = \text{"red"}\} + 1 \\ \max\{\emptyset\} \Rightarrow c_{ij1} = -\infty \\ \max\{c_{hk1} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\} + 1 \\ \max\{\emptyset\} \Rightarrow c_{ij1} = -\infty \end{array} \right.$$

Eq. ricorrenza - Passo ricorsivo per $\text{LCS0red}_v(X_i, Y_j)$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) = \text{"red"}$$

$$\text{LCS0red}_v(X_i, Y_j) = \langle \dots, x_i \rangle$$

$\text{LCS0red}_v(X_i, Y_j)$ non può finire con due elementi consecutivi rossi e non ci devono essere prima altri elementi consecutivi rossi.

$$\text{LCS0red}_v(X_i, Y_j) = \langle \dots, x_p, x_i \rangle = \text{LCS0red}_v(X_p, Y_q) + \langle x_i \rangle$$

$$\text{LCS0red}_v(X_p, Y_q) = \max\{\text{LCS0red}_v(X_h, Y_k) \mid 1 \leq h < i, 1 \leq k < j, \text{col}(x_h) \neq \text{"red"}\}$$

$$\max\{\emptyset\} \Rightarrow \nexists \text{LCS0red}_v(X_p, Y_q) \Rightarrow \text{LCS0red}_v(X_i, Y_j) = \langle x_i \rangle$$

$$c_{ij0} = c_{pq0} + 1$$

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$$c_{ij0} = \max\{c_{hko} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j, \text{col}(x_h) \neq \text{"red"}\} + 1$$
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$$c_{ij0} = \max\{c_{hk0} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j\} + 1$$

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Eq. ricorrenza - Passo ricorsivo per $\text{LCS0red}_v(X_i, Y_j)$

In conclusione, per $\text{LCS0red}_v(X_i, Y_j)$:

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) = \text{"red"}$$

$$c_{ij0} = \max\{c_{hko} \neq -\infty \mid 1 \leq h < i, 1 \leq k < j, \text{col}(x_h) \neq \text{"red"}\} + 1$$
$$\max\{\emptyset\} \Rightarrow c_{ij0} = 1$$

$$1 \leq i \leq m \wedge 1 \leq j \leq n \wedge x_i = y_j \wedge \text{col}(x_i) \neq \text{"red"}$$

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